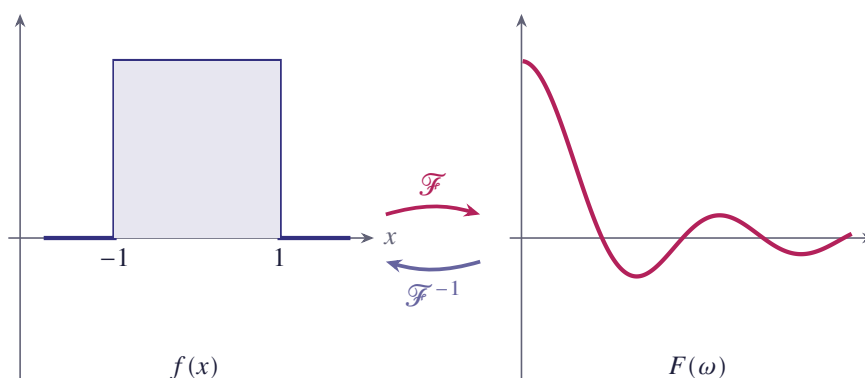


The Fourier Integral

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3.1 Fourier Analysis of Non-Periodic Functions

To be able to use Fourier analysis for non-periodic functions, we first regard the function as periodic and then let the period tend to infinity.

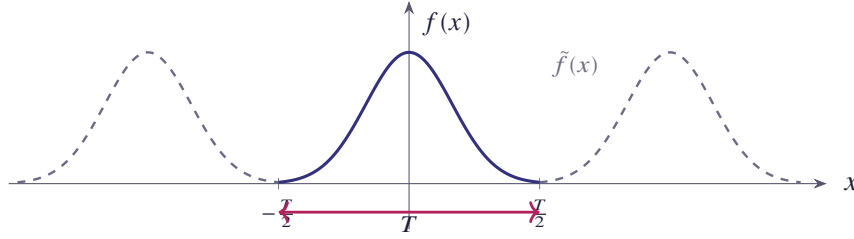


Figure 3.1 A non-periodic function $f(x)$ and its periodic extension $\tilde{f}(x)$ with period T

$$\tilde{f}(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{2\pi}{T} nx + b_n \sin \frac{2\pi}{T} nx$$

$$a_0 = \frac{1}{T} \int_{-T/2}^{T/2} f(x) dx, \quad a_n = \frac{2}{T} \int_{-T/2}^{T/2} f(x) \cos \frac{2\pi}{T} nx dx, \quad b_n = \frac{2}{T} \int_{-T/2}^{T/2} f(x) \sin \frac{2\pi}{T} nx dx$$

$$\begin{aligned} \tilde{f}(x) = & \left(\frac{1}{T} \int_{-T/2}^{T/2} f(t) dt \right) + \sum_{n=1}^{\infty} \left\{ \left(\frac{2}{T} \int_{-T/2}^{T/2} f(t) \cos \frac{2\pi}{T} nt dt \right) \cos \frac{2\pi}{T} nx \right. \\ & \left. + \left(\frac{2}{T} \int_{-T/2}^{T/2} f(t) \sin \frac{2\pi}{T} nt dt \right) \sin \frac{2\pi}{T} nx \right\} \end{aligned}$$

$$\tilde{f}(x) = \left(\frac{1}{T} \int_{-T/2}^{T/2} f(t) dt \right) + \frac{2}{T} \sum_{n=1}^{\infty} \left\{ \int_{-T/2}^{T/2} f(t) \left(\cos \frac{2\pi}{T} nt \cos \frac{2\pi}{T} nx + \sin \frac{2\pi}{T} nt \sin \frac{2\pi}{T} nx \right) dt \right\}$$

$$\tilde{f}(x) = \left(\frac{1}{T} \int_{-T/2}^{T/2} f(t) dt \right) + \frac{2}{T} \sum_{n=1}^{\infty} \left\{ \int_{-T/2}^{T/2} f(t) \cos \frac{2\pi}{T} n(t-x) dt \right\}$$

- To obtain the Fourier analysis of the function $f(x)$ we must let T tend to **infinity**.
- Suppose $f(x)$ is an integrable function, so that

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} f(t) dt = 0$$

Then

$$f(x) = \lim_{T \rightarrow \infty} \frac{2}{T} \sum_{n=1}^{\infty} \left\{ \int_{-T/2}^{T/2} f(t) \cos \frac{2\pi}{T} n(t-x) dt \right\}$$

$$\omega_n \triangleq \frac{2\pi}{T} n \quad \Rightarrow \quad \Delta\omega \triangleq \omega_n - \omega_{n-1} = \frac{2\pi}{T} n - \frac{2\pi}{T} (n-1) = \frac{2\pi}{T} \quad \Rightarrow \quad \frac{2}{T} = \frac{\Delta\omega}{\pi}$$

$$f(x) = \lim_{T \rightarrow \infty} \frac{1}{\pi} \sum_{n=1}^{\infty} \left\{ \int_{-T/2}^{T/2} f(t) \cos \omega_n(t-x) dt \right\} \Delta\omega$$

If $T \rightarrow \infty$, then $\sum_{n=1}^{\infty} \rightarrow \int_0^{\infty}$, $\Delta\omega \rightarrow d\omega$ and $\omega_n \rightarrow \omega$:

$$\begin{aligned} f(x) &= \frac{1}{\pi} \int_0^{\infty} \left\{ \int_{-\infty}^{\infty} f(t) \cos \omega(t-x) dt \right\} d\omega \\ &= \frac{1}{\pi} \int_0^{\infty} \left\{ \int_{-\infty}^{\infty} f(t) (\cos \omega t \cos \omega x + \sin \omega t \sin \omega x) dt \right\} d\omega \\ &= \frac{1}{\pi} \int_0^{\infty} \left\{ \cos \omega x \int_{-\infty}^{\infty} (f(t) \cos \omega t) dt + \sin \omega x \int_{-\infty}^{\infty} (f(t) \sin \omega t) dt \right\} d\omega \end{aligned}$$

$$A(\omega) \triangleq \int_{-\infty}^{\infty} f(t) \cos \omega t \, dt, \quad B(\omega) \triangleq \int_{-\infty}^{\infty} f(t) \sin \omega t \, dt \quad \Rightarrow \quad f(x) \\ = \frac{1}{\pi} \int_0^{\infty} \{A(\omega) \cos \omega x + B(\omega) \sin \omega x\} d\omega$$

3.2 The Fourier Integral

Key result 3.1 ■ The Fourier integral and the sine and cosine transforms

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \{A(\omega) \cos \omega x + B(\omega) \sin \omega x\} d\omega \quad (\text{the Fourier integral; synthesis equation})$$

$$A(\omega) \triangleq \int_{-\infty}^{\infty} f(t) \cos \omega t \, dt \quad (\text{the Fourier cosine transform})$$

$$B(\omega) \triangleq \int_{-\infty}^{\infty} f(t) \sin \omega t \, dt \quad (\text{the Fourier sine transform})$$

(the analysis equations)

Theorem 3.2

A sufficient condition for the existence of the Fourier integral of a function $f(x)$ is that the function be piecewise continuous, have right and left derivatives at every point, and be absolutely integrable, i.e.

$$\int_{-\infty}^{\infty} |f(x)| dx < \infty$$

At points of discontinuity, the value of the Fourier integral equals the average of the left and right limits of $f(x)$ at those points.

3.3 The Complex Form of the Fourier Integral (the Fourier Transform)

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \left\{ \int_{-\infty}^{\infty} f(t) \cos \omega(t-x) dt \right\} d\omega$$

The expression $\int_{-\infty}^{\infty} f(t) \cos \omega(t-x) dt$ is an even function of ω , hence

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left\{ \int_{-\infty}^{\infty} f(t) \cos \omega(t-x) dt \right\} d\omega$$

and the expression $\int_{-\infty}^{\infty} f(t) \sin \omega(t-x) dt$ is an odd function of ω , hence

$$\frac{-i}{2\pi} \int_{-\infty}^{\infty} \left\{ \int_{-\infty}^{\infty} f(t) \sin \omega(t-x) dt \right\} d\omega = 0$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left\{ \int_{-\infty}^{\infty} f(t) \cos \omega(t-x) dt \right\} d\omega - \frac{i}{2\pi} \int_{-\infty}^{\infty} \left\{ \int_{-\infty}^{\infty} f(t) \sin \omega(t-x) dt \right\} d\omega$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left\{ \int_{-\infty}^{\infty} f(t) (\cos \omega(t-x) - i \sin \omega(t-x)) dt \right\} d\omega$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left\{ \int_{-\infty}^{\infty} f(t) e^{-i\omega(t-x)} dt \right\} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left\{ \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \right\} e^{i\omega x} d\omega$$

Key result 3.3 ■ The Fourier transform

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega x} d\omega \quad (\text{the Fourier integral; synthesis equation})$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \quad (\text{the Fourier transform; analysis equation})$$

The complex form of the Fourier series (review):

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{i\frac{2\pi}{T}nx} \quad (\text{Fourier series; synthesis equation}), \quad c_n = \frac{1}{T} \int_T f(x) e^{-i\frac{2\pi}{T}nx} dx$$

(Fourier coefficients; analysis equation)

3.4 The Function Si(u)

$$\text{Si}(x) \triangleq \int_0^x \frac{\sin w}{w} dw$$

- odd;
- has finite asymptotic values at infinity.

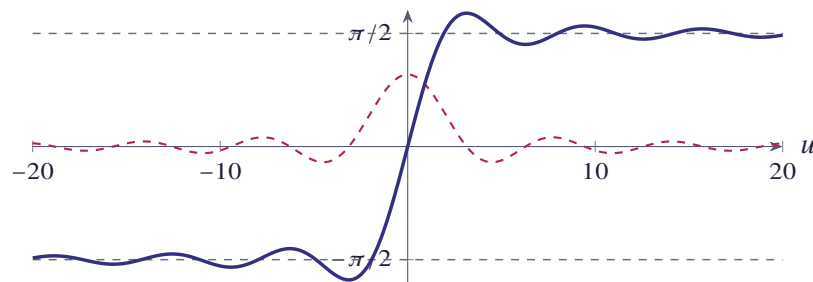


Figure 3.2 The function Si(u) (solid) and the function sin u/u (dashed)

Theorem 3.4 ■ Lemma

$$\int_0^{\infty} \frac{\sin \alpha w}{w} dw = \begin{cases} \frac{\pi}{2} & \alpha > 0 \\ 0 & \alpha = 0 \\ -\frac{\pi}{2} & \alpha < 0 \end{cases}$$

Proof.

For $\alpha = 0$ the lemma is obvious. With the substitution $\tau = \alpha w$:

$$\int_0^x \frac{\sin \alpha w}{w} dw = \int_0^{\alpha x} \frac{\sin \tau}{\tau/\alpha} \frac{d\tau}{\alpha} = \int_0^{\alpha x} \frac{\sin \tau}{\tau} d\tau = \text{Si}(\alpha x), \quad \text{Si}(\alpha x) = -\text{Si}(-\alpha x)$$

So it suffices to prove the lemma for positive values of α .

$$F(\alpha) \triangleq \int_0^\infty e^{-\alpha t} \frac{\sin t}{t} dt, \quad F(0) = \text{Si}(+\infty) = ?$$

$$F'(\alpha) = \int_0^\infty -t e^{-\alpha t} \frac{\sin t}{t} dt = - \int_0^\infty e^{-\alpha t} \sin t dt = - \left[\frac{e^{-\alpha t}}{1 + \alpha^2} (-\alpha \sin t - \cos t) \right]_0^\infty = \frac{-1}{1 + \alpha^2}$$

$$\alpha > 0 : \quad F(\alpha) = -\tan^{-1} \alpha + C, \quad F(\alpha \rightarrow \infty) = 0 \quad \Rightarrow \quad C = F(\infty) + \tan^{-1} \infty = \frac{\pi}{2}$$

$$F(\alpha) = -\tan^{-1} \alpha + \frac{\pi}{2} \quad \Rightarrow \quad F(0) = \text{Si}(+\infty) = \frac{\pi}{2}$$

■

3.5 Examples of the Fourier Integral**Example 3.5**

Express the rectangular pulse below as a Fourier integral.

$$f(x) = \begin{cases} 1 & |x| < 1 \\ 0 & |x| > 1 \end{cases}$$

Since $f(x)$ is even, the Fourier sine transform vanishes: $B(\omega) = 0$.

$$A(\omega) = \int_{-\infty}^\infty f(t) \cos \omega t dt = \int_{-1}^1 \cos \omega t dt = \frac{2}{\omega} \sin \omega$$

$$f(x) = \frac{1}{\pi} \int_0^\infty \frac{2}{\omega} \sin \omega \cos \omega x d\omega = \frac{1}{\pi} \int_0^\infty \frac{1}{\omega} [\sin(1-x)\omega + \sin(1+x)\omega] d\omega$$

$$= \frac{1}{\pi} \left[\int_0^\infty \frac{\sin(1-x)\omega}{\omega} d\omega + \int_0^\infty \frac{\sin(1+x)\omega}{\omega} d\omega \right] = \begin{cases} 0 & x < -1 \\ 1/2 & x = -1 \\ 1 & -1 < x < 1 \\ 1/2 & x = 1 \\ 0 & x > 1 \end{cases}$$

Note: the value of the Fourier integral at the points of discontinuity (using the lemma above).

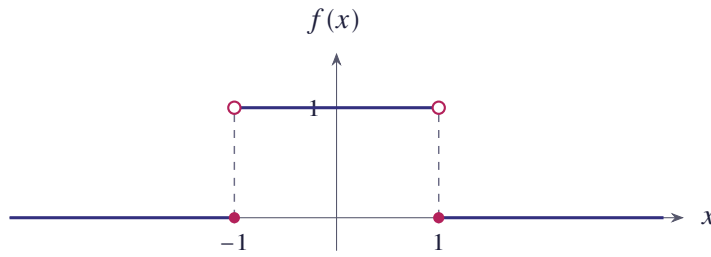


Figure 3.3 The rectangular pulse $f(x) = 1$ for $|x| < 1$

Example 3.6

Express the exponential function below as a Fourier integral.

$$f(x) = \begin{cases} 0 & x < 0 \\ e^{-x} & x > 0 \end{cases}$$

$$\begin{aligned} A(\omega) &= \int_{-\infty}^{\infty} f(x) \cos \omega x \, dx = \int_0^{\infty} e^{-x} \cos \omega x \, dx = \frac{e^{-x}}{1 + \omega^2} (-\cos \omega x + \omega \sin \omega x) \Big|_0^{\infty} \\ &= \frac{1}{1 + \omega^2} \end{aligned}$$

$$\begin{aligned} B(\omega) &= \int_{-\infty}^{\infty} f(x) \sin \omega x \, dx = \int_0^{\infty} e^{-x} \sin \omega x \, dx = \frac{e^{-x}}{1 + \omega^2} (-\sin \omega x - \omega \cos \omega x) \Big|_0^{\infty} \\ &= \frac{\omega}{1 + \omega^2} \end{aligned}$$

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \{A(\omega) \cos \omega x + B(\omega) \sin \omega x\} d\omega = \frac{1}{\pi} \int_0^{\infty} \frac{\cos \omega x + \omega \sin \omega x}{1 + \omega^2} d\omega$$

$$f(x=0) = \frac{1}{\pi} \int_0^{\infty} \frac{1}{1 + \omega^2} d\omega = \frac{1}{\pi} [\tan^{-1} \omega]_0^{\infty} = \frac{1}{\pi} \times \frac{\pi}{2} = \frac{1}{2}$$

Example 3.7

Express the function below as a Fourier integral.

$$f(x) = e^{-|x|}$$

Since $f(x)$ is even, the Fourier sine transform vanishes.

$$\begin{aligned} A(\omega) &= \int_{-\infty}^{\infty} f(x) \cos \omega x \, dx = 2 \int_0^{\infty} e^{-x} \cos \omega x \, dx = \frac{2e^{-x}}{1 + \omega^2} (-\cos \omega x + \omega \sin \omega x) \Big|_0^{\infty} \\ &= \frac{2}{1 + \omega^2} \end{aligned}$$

$$B(\omega) = \int_{-\infty}^{\infty} e^{-|x|} \sin \omega x \, dx = 0$$

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \{A(\omega) \cos \omega x + B(\omega) \sin \omega x\} d\omega = \frac{2}{\pi} \int_0^{\infty} \frac{\cos \omega x}{1 + \omega^2} d\omega$$

$$f(x=0) = \frac{2}{\pi} \int_0^{\infty} \frac{1}{1 + \omega^2} d\omega = \frac{2}{\pi} [\tan^{-1} \omega]_0^{\infty} = \frac{2}{\pi} \times \frac{\pi}{2} = 1$$

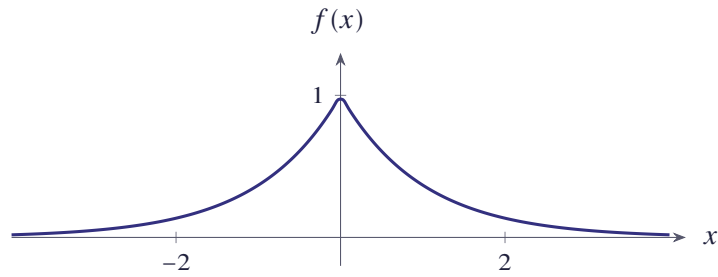


Figure 3.4 The function $f(x) = e^{-|x|}$

Example 3.8

Express the function below as a Fourier integral.

$$f(x) = \begin{cases} \sin x & |x| \leq \pi \\ 0 & |x| > \pi \end{cases}$$

Since $f(x)$ is odd, the Fourier cosine transform vanishes.

$$A(\omega) = \int_{-\pi}^{\pi} \sin x \cos \omega x \, dx = 0$$

$$\begin{aligned} B(\omega) &= \int_{-\pi}^{\pi} \sin x \sin \omega x \, dx = 2 \int_0^{\pi} \sin x \sin \omega x \, dx = \int_0^{\pi} [\cos(1 - \omega)x - \cos(1 + \omega)x] \, dx \\ &= \left[\frac{\sin(1 - \omega)x}{1 - \omega} - \frac{\sin(1 + \omega)x}{1 + \omega} \right]_0^{\pi} = \frac{2 \sin(\omega \pi)}{1 - \omega^2} \end{aligned}$$

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \{A(\omega) \cos \omega x + B(\omega) \sin \omega x\} \, d\omega = \frac{2}{\pi} \int_0^{\infty} \frac{\sin(\omega \pi) \sin \omega x}{1 - \omega^2} \, d\omega$$

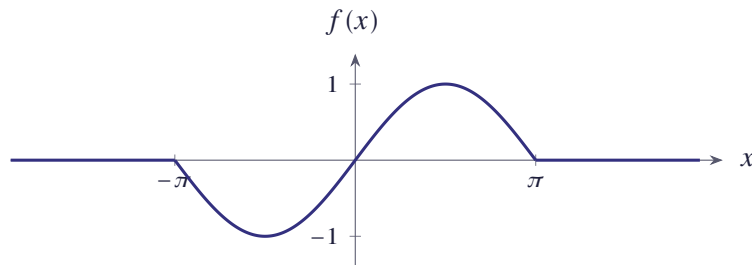


Figure 3.5 The function $f(x) = \sin x$ for $|x| \leq \pi$ and zero outside

Example 3.9

Express the function below as a Fourier integral.

$$f(x) = \begin{cases} x^2 & |x| \leq a \\ 0 & |x| > a \end{cases}$$

Since $f(x)$ is even, the Fourier sine transform vanishes.

$$B(\omega) = \int_{-a}^a x^2 \sin \omega x \, dx = 0$$

$$\begin{aligned} A(\omega) &= \int_{-\infty}^{\infty} f(x) \cos \omega x \, dx = 2 \int_0^a x^2 \cos \omega x \, dx = 2 \left[\frac{x^2}{\omega} \sin \omega x + \frac{2x}{\omega^2} \cos \omega x - \frac{2}{\omega^3} \sin \omega x \right]_0^a \\ &= 2 \left[\left(\frac{a^2}{\omega} - \frac{2}{\omega^3} \right) \sin \omega a + \frac{2a}{\omega^2} \cos \omega a \right] \\ f(x) &= \frac{2}{\pi} \int_0^{\infty} \left[\left(\frac{a^2}{\omega} - \frac{2}{\omega^3} \right) \sin \omega a + \frac{2a}{\omega^2} \cos \omega a \right] \cos \omega x \, d\omega \end{aligned}$$

3.6 Some Properties of the Fourier Integral

All the properties follow from the synthesis and analysis equations:

$$\begin{aligned} f(x) &= \frac{1}{\pi} \int_0^{\infty} \{A(\omega) \cos \omega x + B(\omega) \sin \omega x\} \, d\omega, & A(\omega) &\triangleq \int_{-\infty}^{\infty} f(x) \cos \omega x \, dx, \\ B(\omega) &\triangleq \int_{-\infty}^{\infty} f(x) \sin \omega x \, dx \end{aligned}$$

Time Scaling

$$f(x) \rightarrow f(ax), \quad a \in \mathbb{R} \quad \Rightarrow \quad \begin{cases} A(\omega) \rightarrow \frac{1}{|a|} A\left(\frac{\omega}{a}\right) \\ B(\omega) \rightarrow \frac{1}{|a|} B\left(\frac{\omega}{a}\right) \end{cases}$$

Proof.

$A_1(\omega)$

$$\begin{aligned} &= \int_{-\infty}^{\infty} f(ax) \cos \omega x \, dx \quad \begin{cases} \begin{aligned} &\stackrel{a \geq 0}{u=ax} \int_{-\infty}^{\infty} f(u) \cos \frac{\omega}{a} u \frac{du}{a} = \frac{1}{a} A\left(\frac{\omega}{a}\right) \\ &\stackrel{a \leq 0}{u=ax} \int_{+\infty}^{-\infty} f(u) \cos \frac{\omega}{a} u \frac{du}{a} = -\frac{1}{a} \int_{-\infty}^{\infty} f(u) \cos \frac{\omega}{a} u \, du = -\frac{1}{a} A\left(\frac{\omega}{a}\right) \end{aligned} \end{cases} \\ &\Rightarrow A_1(\omega) = \frac{1}{|a|} A\left(\frac{\omega}{a}\right) \end{aligned}$$

■

Parseval's Identity

$$\int_{-\infty}^{\infty} f^2(x) dx = \frac{1}{\pi} \int_0^{\infty} \{A^2(\omega) + B^2(\omega)\} d\omega$$

Proof.

$$\begin{aligned} \int_{-\infty}^{\infty} f^2(x) dx &= \int_{-\infty}^{\infty} f(x) \left(\frac{1}{\pi} \int_0^{\infty} \{A(\omega) \cos \omega x + B(\omega) \sin \omega x\} d\omega \right) dx \\ &= \frac{1}{\pi} \int_0^{\infty} \left\{ A(\omega) \left[\int_{-\infty}^{\infty} f(x) \cos \omega x dx \right] + B(\omega) \left[\int_{-\infty}^{\infty} f(x) \sin \omega x dx \right] \right\} d\omega \\ &= \frac{1}{\pi} \int_0^{\infty} \{A^2(\omega) + B^2(\omega)\} d\omega \end{aligned}$$

■

The Time Derivative

$$\begin{aligned} f(x) &= \frac{1}{\pi} \int_0^{\infty} \{A(\omega) \cos \omega x + B(\omega) \sin \omega x\} d\omega \implies f'(x) \\ &= \frac{1}{\pi} \int_0^{\infty} \{\omega B(\omega) \cos \omega x - \omega A(\omega) \sin \omega x\} d\omega \end{aligned}$$

(using the rule for differentiating an integral:)

$$\frac{\partial}{\partial z} \int_{a(z)}^{b(z)} f(x, z) dx = \int_{a(z)}^{b(z)} \frac{\partial f(x, z)}{\partial z} dx + f(b(z), z) \frac{db(z)}{dz} - f(a(z), z) \frac{da(z)}{dz}$$

The Frequency Derivative

$$\frac{dA(\omega)}{d\omega} = A'(\omega) = \int_{-\infty}^{\infty} -xf(x) \sin \omega x dx \implies \int_{-\infty}^{\infty} (xf(x)) \sin \omega x dx = -A'(\omega)$$

$$\frac{dB(\omega)}{d\omega} = B'(\omega) = \int_{-\infty}^{\infty} xf(x) \cos \omega x dx \implies \int_{-\infty}^{\infty} (xf(x)) \cos \omega x dx = B'(\omega)$$

$$xf(x) = \frac{1}{\pi} \int_0^{\infty} \{B'(\omega) \cos \omega x - A'(\omega) \sin \omega x\} d\omega$$

Exercise 3.10

Prove the following identities:

1. $\int_0^\infty \frac{\cos xw + w \sin xw}{1 + w^2} dw = \begin{cases} 0 & x < 0 \\ \pi/2 & x = 0 \\ \pi e^{-x} & x > 0 \end{cases}$
2. $\int_0^\infty \frac{\sin \pi w \sin xw}{1 - w^2} dw = \begin{cases} \frac{\pi}{2} \sin x & 0 \leq x \leq \pi \\ 0 & x > \pi \end{cases}$
3. $\int_0^\infty \frac{1 - \cos \pi w}{w} \sin xw dw = \begin{cases} \frac{1}{2} \pi & 0 < x < \pi \\ 0 & x > \pi \end{cases}$
4. $\int_0^\infty \frac{\cos \frac{1}{2} \pi w}{1 - w^2} \cos xw dw = \begin{cases} \frac{1}{2} \pi \cos x & 0 < |x| < \frac{1}{2} \pi \\ 0 & |x| \geq \frac{1}{2} \pi \end{cases}$
5. $\int_0^\infty \frac{\sin w - w \cos w}{w^2} \sin xw dw = \begin{cases} \frac{1}{2} \pi x & 0 < x < 1 \\ \frac{1}{4} \pi & x = 1 \\ 0 & x > 1 \end{cases}$